

# Obidi's Metaphysics

*Axiomatic System: The Axioms of Entropicity*

---

**Author:** John Onimisi Obidi (INFINITE)

**System:** Theory of Entropicity (ToE)

**Edition:** First Formal Axiomatic Edition

**Date:** June 10, 2026

**Classification:** Metaphysics — Formal Axiomatic Philosophy

**Location:** Regina, Saskatchewan, Canada

---

## Methodological Note

This document follows the tradition of formal axiomatic systems in the history of philosophy and foundational science — after the manner of Euclid's *Elements*, Newton's *Principia Mathematica*, Spinoza's *Ethics*, and Kolmogorov's probability axioms — and applies that tradition rigorously to the domain of metaphysics. The system proceeds from primitive (undefined) concepts, through formal definitions, axioms, postulates, theorems, and corollaries. No theorem is asserted without derivation from the axioms. No axiom is introduced without philosophical justification.

## PREAMBLE

---

## Purpose of This Document

The purpose of this document is to lay the formal axiomatic foundations of **Obidi's Metaphysics** — also designated the **Theory of Entropicity (ToE)** — as a rigorous, self-consistent system from which all metaphysical theorems and ontological claims of the theory may be derived with logical precision. The axioms set forth herein constitute the irreducible commitments of the system: they are not themselves derived within the system, but are the bedrock from which all derivation proceeds.

Entropicity is a metaphysical framework that identifies the *Entropic Field* as the ontological ground of all being, information as the primitive substance of reality, and entropy as the measure of ontological potentiality. The system makes claims about the nature of existence, causality, time, space, mind, identity, emergence, and value — all grounded in the dynamics of a single unified Entropic Field governed by the Obidi Action.

## Authorial Note

*I undertook the axiomatization of this system because philosophy without rigor is mere speculation, and speculation without accountability to formal structure produces no cumulative knowledge. The history of metaphysics is littered with brilliant intuitions undone by conceptual imprecision. If Entropicity is to be more than an intellectual gesture — if it is to constitute genuine philosophical progress — then it must submit itself to the same discipline that has made mathematics and logic the most reliable instruments of thought humanity has produced. These axioms are not claimed to be self-evidently true in the Cartesian sense; they are claimed to be the most economical and explanatorily powerful starting points for a comprehensive metaphysical system. They are offered as a living framework: open to refinement, extension, and challenge.*

## Structure of the System

The axiomatic system of Entropicity is organized according to the following methodological hierarchy:

1. **Primitive Concepts** — undefined terms that serve as the ultimate basis of all definitions.
2. **Definitions (D1-D15)** — formally defined terms constructed from primitive concepts and prior definitions.
3. **Axioms (A1-A12)** — foundational assertions accepted without proof within the system.
4. **Postulates (P1-P6)** — domain-specific assumptions that extend the axioms to particular areas of inquiry.
5. **Theorems (T1-T10)** — propositions derived by logical inference from axioms, postulates, and prior theorems.
6. **Corollaries (C1-C8)** — propositions following immediately from theorems with minimal additional argument.

## PART I — PRIMITIVE CONCEPTS (UNDEFINED TERMS)

---

*The following six concepts are taken as primitive: they are not defined in terms of more basic notions within this system, but are instead grasped through their roles within the axiomatic structure and through pre-theoretical philosophical intuition. Their formal behavior is fixed entirely by the axioms.*

## I.1 The Entropic Field $\varphi_E$

The **Entropic Field**, denoted  $\varphi_E$ , is the ontological medium of all being. It is not a physical field in the ordinary sense, nor a Platonic form, nor a substance in the Aristotelian sense. It is the primitive ground from which all particular entities, processes, and relations emerge.  $\varphi_E$  is taken as ontologically prior to space, time, matter, and mind — these are derivative structures of the field. It is continuous, total, and inexhaustible.

## I.2 Information $\mathcal{I}$

**Information**, denoted  $\mathcal{I}$ , is the fundamental currency of ontological structure. It is not reducible to physical bits, symbolic representations, or epistemic content, though it underlies all of these. In the context of Entropicity,  $\mathcal{I}$  is the primitive differentiating factor: that by virtue of which any state of  $\varphi_E$  is *this* state and not another. Information is the measure of ontological determinacy.

## I.3 Entropy $S$

**Entropy**, denoted  $S$ , is the measure of informational potentiality within a region or state of the Entropic Field. In Entropicity, entropy is not merely a thermodynamic quantity; it is the metaphysically fundamental measure of possibility-space. High entropy corresponds to high ontological potentiality; low entropy to high ontological determinacy. Entropy and information stand in a complementary, not merely inverse, relation.

## I.4 Becoming $\mathfrak{B}$

**Becoming**, denoted  $\mathfrak{B}$ , is the primitive ontological process by which being actualizes. It is not mere change in the Humean sense (succession of states), but the irreversible actualization of potential through the dynamics

of  $\varphi_E$ . Becoming is ontologically more fundamental than Being understood as static presence; being is a moment within Becoming, not vice versa.

## I.5 The Entropic Subject $\Sigma$

The **Entropic Subject**, denoted  $\Sigma$ , is any self-organizing informational node within  $\varphi_E$  that possesses the capacity for self-reference and internal modeling. Entropic Subjects are not limited to conscious entities; any sufficiently complex self-organizing structure qualifies, though conscious minds represent high-order instances.  $\Sigma$  is a primitive in that its self-organizing character cannot be fully captured by reference to lower-level components alone.

## I.6 The Obidi Action $\mathcal{P}_{\text{Obidi}}$

The **Obidi Action**, denoted  $\mathcal{P}_{\text{Obidi}}$ , is the fundamental operator governing entropic actualization across  $\varphi_E$ . Analogous in structure to the action functionals of classical and quantum mechanics, but operating at the ontological level,  $\mathcal{P}_{\text{Obidi}}$  selects the trajectory through possibility-space that reality actualizes. It is the primitive law-like structure of the Entropic Field, prior to all particular physical laws, which are its domain-specific expressions.

# PART II — DEFINITIONS

---

*The following definitions are formal. Each definition (Dn) introduces a term by specifying necessary and sufficient conditions in terms of primitive concepts and previously introduced definitions. Definitions are not substantive claims; they are conceptual legislation.*

**D1. Being.** Any state  $\omega$  of the Entropic Field  $\varphi_E$  such that its informational content satisfies  $\mathcal{I}(\omega) > 0$ . Being is thus constitutively informational; it is not a bare substratum but a determinate informational configuration of  $\varphi_E$ .

---

**D2. Non-Being.** The limiting state of  $\varphi_E$  approached as  $\mathcal{I} \rightarrow 0$ . Non-Being is an asymptotic limit: it is never fully realized within a field that is (by A1) ontologically prior and inexhaustible. Non-Being cannot be an actual state of  $\varphi_E$ ; it is a conceptual boundary condition.

---

**D3. Entropic Particular.** Any entity  $E$  defined by its unique informational-entropic trajectory  $\tau(E) = \{\omega_t\}$  through  $\varphi_E$  across entropic intervals. An Entropic Particular is not a substance but a trajectory; its identity is constituted by the coherence of this trajectory, not by the persistence of an underlying material substrate.

---

**D4. Relation.** A structural dependency  $R(E_1, E_2, \dots E_n)$  between two or more Entropic Particulars, mediated by the structure of  $\varphi_E$ . Relations are not mere logical or conceptual connections; they are real features of the informational geometry of the Entropic Field. No Entropic Particular exists independently of its relations.

---

**D5. Causality.** An asymmetric informational dependency relation. An Entropic Particular  $A$  causes  $B$  if and only if (i)  $\mathcal{I}(A)$  constrains the possible states of  $\mathcal{I}(B)$ , and (ii)  $A$  is prior to  $B$  in entropic order. Causality is thus grounded in informational structure, not in Humean constant conjunction or Kantian categorical imposition.

---

**D6. Time.** The directed total ordering of entropic states of  $\varphi_E$ . Time  $t$  increases in the direction of entropy increase within the field. Time is not a container or background dimension; it is a derived structure of entropic Becoming  $\mathfrak{B}$ . There is no time independent of entropic process.

---

**D7. Space.** The informational geometry of the Entropic Field  $\varphi_E$ : the manifold structure induced by the distribution and gradients of entropy across the field. Space is the relational structure of co-present informational configurations; it is not an absolute container (contra Newton) nor a purely formal intuition (contra Kant), but an emergent geometric property of  $\varphi_E$ .

---

**D8. Mind / Consciousness.** A high-order self-referential entropic structure  $\Sigma_M$  within  $\varphi_E$  that possesses the capacity to model the field itself — including, reflexively, its own modeling activity. Consciousness is not a separate substance (contra Descartes) nor an illusion (contra eliminative materialism); it is a genuine, causally efficacious, emergent level of the Entropic Field.

---

**D9. Identity.** The persistence of an Entropic Particular's informational trajectory  $\tau(E)$  across entropic intervals in a manner that preserves sufficient informational coherence to individuate  $E$  across time. Identity is a matter of degree; it admits of gradation along the axis of informational continuity.

---

**D10. Emergence.** The formation of a higher-order Entropic Particular  $E_H$  from the collective dynamics of lower-order ones  $\{E_i\}$  within  $\varphi_E$ , such that  $E_H$  possesses properties  $P$  that are not possessed by any  $E_i$  individually, nor reducible to mere aggregation of the  $\{E_i\}$ . Emergence is ontologically real, not merely epistemic.

---

**D11. Value.** A measure  $V$  of informational coherence and structural complexity within an Entropic Subject  $\Sigma$ . Value is not a merely subjective projection; it is grounded in the objective informational dynamics of  $\varphi_E$ . Higher informational coherence within an Entropic Subject corresponds to greater value, with ethical and aesthetic values as domain-specific instances.

---

**D12. Necessity.** A relation, property, or state that holds across all possible configurations of  $\varphi_E$ . Necessity in Entropicity is not de dicto logical necessity (truth in all possible worlds simpliciter) but de re entropic necessity: invariance across the total possibility-space of the Entropic Field.

---

**D13. Contingency.** A relation, property, or state that holds in some but not all configurations of  $\varphi_E$ . Contingent features of reality are those whose informational content is not entailed by the axioms and structure of the Entropic Field alone, but depends on particular entropic trajectories that might have been otherwise.

---

**D14. The Entropic Ground.** The total state of  $\varphi_E$  considered as the undivided informational plenum from which all particular beings emerge. The Entropic Ground is not itself an Entropic Particular; it is the field as a whole, prior to differentiation into discrete entities. It is the closest analogue within Entropicity to what classical metaphysics called *Being qua Being*.

---

**D15. Actualization.** The process by which a potential entropic state  $\omega_p$  becomes a determinate Entropic Particular  $E$  through the operation of the Obidi Action  $\mathcal{P}_{\text{Obidi}}$  upon  $\varphi_E$ . Actualization is the transition from entropic potentiality to ontological determinacy; it is the fundamental event of the metaphysical system.

---

# PART III — AXIOMS OF ENTROPICITY

## (A1-A12)

---

*The twelve Axioms of Entropicity constitute the irreducible commitments of the system. They are not empirical generalizations, nor analytic truths, but synthetic foundational posits adopted for their explanatory power, coherence, and adequacy to the full range of ontological phenomena.*

### A1 — Axiom of the Ground

*$\varphi_E$  exists, and for all Entropic Particulars  $E$ :  $E$  is ontologically posterior to  $\varphi_E$ .*

**Gloss:** The Entropic Field exists and is ontologically prior to all particular beings. No particular entity is more fundamental than the field from which it emerges.

**Commentary:** *This axiom performs the role of the Principle of Sufficient Ground in classical metaphysics, but without postulating a theological grounding entity. The Entropic Field is not God, Substance (Spinoza), nor the Absolute (Hegel) — it is the minimally posited ontological ground consistent with the rest of the system. The claim of ontological priority means that the existence and nature of any particular being presupposes the field, but not vice versa.*

### A2 — Axiom of Informational Primacy

*For all states  $\omega$  of  $\varphi_E$ :  $\omega$  is constituted by  $\mathcal{I}(\omega)$ .  $\mathcal{I}$  is the ontological primitive.*

**Gloss:** All being is constituted by information. Information is the fundamental substance of reality — not matter, not energy, not mind as traditionally

conceived.

**Commentary:** *This axiom places Entropicity within the tradition of information-theoretic ontologies, resonating with John Archibald Wheeler's "It from Bit" conjecture, but radicalizing it into a full metaphysical commitment. It is not claimed that physical information (bits, qubits) exhausts the concept; rather, information in the metaphysical sense — as pure ontological determinacy — is the primitive from which physical, mental, and social reality are all derived.*

### A3 — Axiom of Entropic Becoming

*For all  $E$  in  $\varphi_E$ :  $E$  undergoes  $\mathcal{B}$  such that  $dS(E)/dt \geq 0$ ; equality holds only at ideal limits.*

**Gloss:** No being is fully static. All being is a process of Becoming structured by the non-decrease of entropy. Being is inherently processual.

**Commentary:** *This axiom aligns Entropicity with the Heraclitean and Whiteheadian process traditions, while grounding the processual character of reality in the precise, quantifiable notion of entropy increase. The apparent stability of persistent objects (stones, mountains, persons) is understood as a high-information, low-entropy trajectory within Becoming — a pattern of change, not an absence of it. At ideal limits (zero Kelvin, theoretical closed systems), the rate of Becoming approaches zero but never reaches it within the actual field.*

### A4 — Axiom of the Obidi Action

*All entropic actualization is governed by  $\mathcal{S}_{Obidi}$ ; the actual trajectory of  $\varphi_E$  is that which extremizes  $\mathcal{S}_{Obidi}$  over the total configuration space of  $\varphi_E$ .*

**Gloss:** Reality follows the path that extremizes the Obidi Action over the Entropic Field. The Obidi Action is the fundamental dynamical law of

metaphysical reality.

**Commentary:** *Action principles occupy a privileged position in physics — from Lagrangian mechanics to quantum field theory, the principle of stationary action is arguably the deepest structural feature of physical law. The Obidi Action generalizes this principle to the ontological level: it is the master law from which all domain-specific laws (physical, biological, cognitive) are derivable as constrained instances. The extremization condition (rather than minimization) reflects the richer possibility-space of a metaphysical, rather than purely physical, action principle.*

## A5 — Axiom of Irreversibility

*For all processes  $P$  in  $\varphi_E$ :  $\mathcal{B}(P)$  is irreversible; there is no entropic trajectory that exactly reverses  $P$ .*

**Gloss:** The direction of Becoming is irreversible. Entropic order defines the asymmetry of time. The past is ontologically fixed; the future is ontologically open.

**Commentary:** *This axiom takes the thermodynamic arrow of time and elevates it from a statistical tendency to an ontological necessity. The irreversibility of Becoming is not a merely practical limitation (insufficient information to reverse) but a fundamental feature of the structure of  $\varphi_E$ . It grounds the asymmetry of causality (D5), the directionality of time (D6), and the ontological reality of the past as distinct from the future. It directly challenges time-symmetric interpretations of fundamental physics by locating temporal asymmetry at the metaphysical rather than the merely thermodynamic level.*

## A6 — Axiom of Relational Ontology

*For all  $E$  in  $\varphi_E$ : the properties of  $E$  are constituted by  $R(E, \{E_{ij}\})$ , not by intrinsic properties of  $E$  in isolation.*

**Gloss:** Relations are ontologically primary. Entropic Particulars are constituted by their relational positions within the Entropic Field, not by intrinsic, self-subsistent properties.

**Commentary:** *This axiom allies Entropicity with the relational ontologies of Leibniz (in his anti-Newtonian moments), contemporary structural realism (Ladyman, Ross), and quantum mechanical holism. What a thing **is** is inseparable from how it stands with respect to the rest of the Entropic Field. There are no ontological atoms: no entities whose nature is fully determined prior to and independently of their relations. The Entropic Particular (D3) is thus not a substance that **enters into** relations, but a node **constituted by** relations.*

## **A7 — Axiom of Informational Geometry**

*$\varphi_E$  possesses a geometric structure  $\mathbb{G}(\varphi_E)$  determined by the manifold of Shannon entropy distributions;  $\mathbb{G}(\varphi_E)$  is the topology of reality.*

**Gloss:** The Entropic Field has a genuine geometric structure determined by the distribution of entropy across it. This geometry is not merely mathematical; it is the real topology of ontological reality.

**Commentary:** *The application of information geometry (developed by Amari and others in the context of statistical manifolds) to ontology is one of the distinctive technical contributions of Entropicity. The curvature of ontological space-time is not (fundamentally) Riemannian curvature induced by mass-energy (as in General Relativity), but entropic curvature induced by information-entropy gradients. General Relativity is recovered as a domain-specific approximation at the physical level — what the system calls Statistical Gravity.*

## A8 — Axiom of Emergence

*For all collections  $\{E_i\}$  in  $\varphi_E$ : there exist higher-order Entropic Particulars  $E_H$  such that  $P(E_H) \notin \bigcup P(E_i)$  — irreducibly emergent properties are real.*

**Gloss:** Higher-order Entropic Particulars emerge irreducibly from the collective dynamics of lower-order ones. Emergent properties are genuine ontological features, not merely epistemic simplifications.

**Commentary:** *This axiom commits Entropicity to strong (ontological) emergence, in contrast to weak (epistemological) emergence. It is the metaphysical foundation for the irreducibility of biology to chemistry, mind to neuroscience, and social reality to individual psychology. The axiom does not preclude bottom-up causal influences; it asserts that top-down causal powers are equally real, a position sometimes called "downward causation." The formal condition states that emergent properties fall outside the mere union of lower-level properties — they arise from the collective informational dynamics in a manner that the system's axioms guarantee to be ontologically novel.*

## A9 — Axiom of the Entropic Subject

*$\exists \Sigma$  in  $\varphi_E$  such that  $\Sigma$  models  $\varphi_E$  and is a genuine ontological node:  $\Sigma$  is not epiphenomenal.*

**Gloss:** There exists at least one self-referential Entropic Subject capable of modeling the Entropic Field. Such subjects are genuine ontological nodes with causal efficacy, not mere byproducts of physical processes.

**Commentary:** *This axiom secures the ontological reality of mind against eliminativist and epiphenomenalist reductions. It does not assert that all possible configurations of  $\varphi_E$  contain an Entropic Subject; it asserts only that some do — and that when they do, those subjects are causally real. The existential quantifier*

*is backed by the evident fact of the present inquiry: this very axiomatic system, being produced by an Entropic Subject, demonstrates the existence of at least one. The axiom also grounds the philosophical possibility of Entropicity's own formulation.*

#### **A10 — Axiom of Ontological Continuity**

*$\varphi_E$  is continuous: for all  $E_1, E_2$  in  $\varphi_E$ , there exists a continuous entropic path between them; there are no ontological gaps.*

**Gloss:** The Entropic Field is continuous. There are no ontological gaps, voids, or discontinuities in being — only gradients of entropy. Reality is seamless.

**Commentary:** *This axiom rules out ontological dualism at the fundamental level: there is no gap between mind and matter, between the physical and the mental, between the natural and the normative, because all such apparent gaps are, at the level of  $\varphi_E$ , merely entropic gradients within a continuous field. The apparent discreteness of reality (particles, organisms, persons, social institutions) is emergent structure within a fundamentally continuous ontological medium. This does not preclude quantum discreteness at the physical level, which is recovered as a domain-specific structure within the continuous Entropic Field.*

#### **A11 — Axiom of Causal Closure**

*For all causal relations  $C$  in  $\varphi_E$ :  $C$  is an informational-entropic relation within  $\varphi_E$ ;  $\forall x: x$  causes  $y$  only if  $x, y \in \varphi_E$ .*

**Gloss:** All causal relations are informational-entropic relations within the Entropic Field. There are no causal influences external to the field — no

supernatural, transcendent, or extra-ontological causes.

**Commentary:** *This axiom is analogous to the Principle of Causal Closure accepted in contemporary philosophy of mind, but generalized to the metaphysical level. It does not deny the reality of intentional, mental, or normative causation; it asserts that all such causation operates through informational-entropic mechanisms within  $\varphi_E$ . The axiom has implications for theology and supernatural metaphysics: if any transcendent being acts causally within reality, that being and its action must fall within the scope of  $\varphi_E$  — otherwise the causal influence is unintelligible.*

## A12 — Axiom of Ontological Sufficiency

$\varphi_E$  together with  $\mathcal{S}_{Obidi}$  and  $\{A1, ..., A12\}$  is sufficient to ground all metaphysical facts:  $\forall F_{meta}: F_{meta}$  is derivable from  $(\varphi_E, \mathcal{S}_{Obidi}, \{A_n\})$ .

**Gloss:** The Entropic Field, together with the Obidi Action and the Axioms of Entropicity, is sufficient to ground all metaphysical facts. No additional ontological posits are required.

**Commentary:** *This is the principle of ontological parsimony for the system as a whole. It does not claim that the system can currently derive all metaphysical facts explicitly — the Open Problems section (Part VII) acknowledges significant gaps. It claims that in principle, the resources of the system are sufficient: any metaphysical fact that obtains does so in virtue of the structure of  $\varphi_E$  as governed by  $\mathcal{S}_{Obidi}$  and the axioms. The axiom is a commitment to explanatory completeness as a regulatory ideal.*

## PART IV — POSTULATES (P1-P6)

---

*Postulates extend the axioms to specific domains of inquiry. Unlike axioms, postulates make more concrete theoretical commitments that are in principle revisable in light of developments in specific sciences or further philosophical analysis. They are domain-specific rather than maximally general.*

### **P1 — The Informational Postulate**

Shannon entropy  $H = -\sum p_i \log p_i$  is the correct formal measure of ontological potentiality within  $\varphi_E$ . The information-theoretic framework of Shannon (1948) is adopted as the operative formalism for quantifying the informational structure of being. Extensions to quantum information (von Neumann entropy) are recognized as necessary at the quantum level.

### **P2 — The Temporal Postulate**

The arrow of time is identical to the direction of entropy increase in  $\varphi_E$ . The felt asymmetry of time — the sense in which the past is fixed and the future open, in which memory is of the past and anticipation of the future — is grounded in, and identical to, the directional entropic structure of the Entropic Field. The psychological arrow, the cosmological arrow, and the thermodynamic arrow of time are all aspects of the single ontological arrow defined by Becoming  $\mathfrak{B}$ .

### **P3 — The Consciousness Postulate**

Consciousness is constituted by high-order self-referential informational structures within  $\varphi_E$  and is causally efficacious. Specifically: a system is

conscious to the degree that it integrates information in a self-referential manner across the Entropic Field — resonating with Integrated Information Theory (Tononi) but grounding it in the metaphysics of Entropicity rather than in neuroscience. Consciousness is not emergent upon the physical level alone but upon the informational structure of  $\varphi_E$ .

#### **P4 — The Geometric Postulate**

The geometry of  $\varphi_E$  is non-Euclidean; its curvature is induced by informational-entropy gradients, constituting what the system designates *Statistical Gravity*. Physical gravitational fields are the domain-specific expression of the curvature of ontological geometry induced by high-density informational-entropic configurations within  $\varphi_E$ . General Relativity is an approximation valid at the physical level; the more fundamental description is informational-geometric.

#### **P5 — The Emergence Postulate**

Biological, psychological, and social reality are emergent levels of the Entropic Field, each with irreducible ontological properties not fully expressible in the vocabulary of lower-level descriptions. Life, mind, and society are real — they are not illusions to be "explained away" by physics or neuroscience — but neither are they ontologically separate from  $\varphi_E$ . They are the field organizing itself at successively higher levels of informational complexity.

### **P6 — The Value Postulate**

Value — ethical, aesthetic, and epistemic — is a measure of informational coherence and complexity within the dynamics of  $\varphi_E$ , and is therefore objective in the same sense as other structural features of the field. Value facts are not merely projected by Entropic Subjects upon a value-neutral field; they are features of the field as organized by high-order Entropic Subjects. The field itself, through the emergence of complex Entropic Subjects capable of valuing, generates genuine normative structure.

## **PART V — THEOREMS OF ENTROPICITY (T1-T10)**

---

*Each theorem is a proposition derivable by valid inference from the axioms, postulates, and prior theorems of the system. Proof sketches indicate the logical structure of derivation; full formal proofs require the modal and formal logical apparatus of the extended system.*

### **T1 — The Impossibility of Absolute Non-Being**

*Theorem: There is no possible configuration of  $\varphi_E$  in which  $\mathcal{I}(\omega) = 0$  exactly. Absolute Non-Being is impossible.*

#### **Proof Sketch:**

By A1,  $\varphi_E$  exists and is ontologically prior to all particulars. By A2, all states of  $\varphi_E$  are constituted by  $\mathcal{I}$ . By D2, Non-Being is the limit as  $\mathcal{I} \rightarrow 0$  — an asymptotic condition. Since  $\varphi_E$  exists (A1) and is characterized by information (A2), any state of the field has informational content greater than zero. The limit  $\mathcal{I} = 0$  is never

achieved within an existent field. Therefore, absolute Non-Being cannot be a realized state of  $\varphi_E$ . QED.

*Philosophical Significance: T1 establishes the ultimate persistence of being and provides the metaphysical basis for the ancient claim that "ex nihilo nihil fit" (from nothing, nothing comes) — but reconceived informationally: information cannot be absolutely destroyed, only transformed.*

## **T2 — The Uniqueness of the Entropic Ground**

*Theorem: There is exactly one Entropic Ground. The Entropic Field  $\varphi_E$  is numerically unique.*

### **Proof Sketch:**

Suppose there were two distinct Entropic Fields  $\varphi_E^1$  and  $\varphi_E^2$ . By A1, each would be ontologically prior to all particulars. By A11 (Causal Closure), all causal relations are internal to  $\varphi_E$ . But if two fields exist, there can be no causal or relational connection between them (by A11 applied to each), making them mutually inaccessible and ontologically unintelligible relative to each other. By A12 (Sufficiency), the system requires no posit beyond a single field. By parsimony consistent with A12, and by the incoherence of causal isolation between putative grounds, there is exactly one Entropic Ground. QED.

*Philosophical Significance: T2 rules out ontological pluralism at the level of grounds (though it is fully compatible with plurality at the level of Entropic Particulars). There is one reality, one ontological field.*

## **T3 — The Asymmetry of Causality**

*Theorem: Causality in  $\varphi_E$  is necessarily asymmetric: if A causes B, then*

*B does not cause A.*

**Proof Sketch:**

By D5, A causes B iff (i)  $\mathcal{J}(\mathbf{A})$  constrains possible states of  $\mathcal{J}(\mathbf{B})$ , and (ii) A is prior to B in entropic order. By A5 (Irreversibility), entropic order is irreversible: if A is prior to B in entropic order, B is not prior to A. Therefore the asymmetric priority condition in D5(ii) cannot be simultaneously satisfied by A for B and B for A.

Causal asymmetry follows directly. QED.

*Philosophical Significance: T3 provides a metaphysical grounding for the asymmetry of causation, resolving the problem of causal direction without appeal to human cognitive categories (contra Kant) or bare Humean regularity.*

**T4 — The Emergence of Time from Entropy**

*Theorem: Time is a derived structure of the Entropic Field; it does not exist independently of entropic Becoming.*

**Proof Sketch:**

By D6, time is the directed ordering of entropic states. By A3, all being undergoes Becoming structured by entropy increase. By A5, Becoming is irreversible, generating a directed ordering. By P2, the arrow of time is identical to the direction of entropy increase. Therefore, time exists as and only as the directional structure of Becoming in  $\varphi_E$ ; there is no time external to or independent of this entropic process. In a hypothetical field with no entropy change, there would be no time. QED.

*Philosophical Significance: T4 dissolves the ancient problem of time's relation to change by identifying time with the directional entropic structure of the Entropic Field. It is consistent with, and provides a deeper grounding for, relational theories of time (Leibniz, Mach, Barbour).*

## **T5 — The Relational Constitution of Identity**

*Theorem: The identity of any Entropic Particular  $E$  is constituted by its relational trajectory within  $\varphi_E$ , not by any intrinsic, non-relational property.*

### **Proof Sketch:**

By D3, an Entropic Particular is defined by its informational-entropic trajectory. By A6, all properties of any entity are constituted by its relations within  $\varphi_E$ . By D9, identity is the persistence of informational trajectory with sufficient coherence. Therefore, the identity of  $E$  is fully determined by its relational trajectory; no intrinsic, relation-independent properties individuate  $E$ . There are no "bare particulars" in Entropicity. QED.

*Philosophical Significance: T5 provides the metaphysical basis for Corollaries C1 (substance as abstraction) and C2 (personal identity as trajectory), dissolving the puzzles of personal identity generated by substance-attribute metaphysics.*

## **T6 — The Emergence of Consciousness**

*Theorem: Consciousness is a real, irreducibly emergent, causally efficacious property of the Entropic Field at sufficiently high levels of self-referential informational organization.*

### **Proof Sketch:**

By A8, irreducibly emergent properties are real. By A9, there exists at least one Entropic Subject  $\Sigma$  that models  $\varphi_E$  and is a genuine ontological node. By P3, consciousness is constituted by high-order self-referential informational structures within  $\varphi_E$  and is causally efficacious. By D8, consciousness is precisely such a self-

referential structure. By A8 applied to the specific case of high-order self-referential informational organization, the emergent properties of consciousness are irreducible. The conjunction of these results establishes T6. QED.

*Philosophical Significance: T6 resolves the "hard problem of consciousness" within Entropicity not by dissolving it but by locating consciousness as a genuine ontological level of the field. Consciousness is neither a mystery to be explained away nor a separate substance to be duplicated.*

## **T7 — The Ontological Status of Information**

*Theorem: Information is ontologically fundamental — it is neither reducible to matter nor to mind, but is the common ground of both.*

### **Proof Sketch:**

By A2,  $\mathcal{I}$  is the ontological primitive. By A12, the system is ontologically sufficient with  $\varphi_E$  and  $\mathcal{I}_{\text{Obidi}}$  as grounding. Matter (physical states) and mind (D8) are both constituted by informational structures within  $\varphi_E$ . Since information is primitive and constitutive of both, and by T2 there is only one field, information is the common ontological ground of matter and mind: neither is reducible to the other because both are expressions of information in different organizational modes. QED.

*Philosophical Significance: T7 places Entropicity in the tradition of dual-aspect monisms (Spinoza, Chalmers' naturalistic dualism) while going beyond them by grounding both aspects in the single primitive of information rather than in a neutral substance.*

## **T8 — The Impossibility of Ontological Gaps**

*Theorem: There are no ontological gaps within  $\varphi_E$ ; apparent discontinuities are entropic gradients within a continuous field.*

**Proof Sketch:**

By A10,  $\varphi_E$  is continuous. By A11, all causal relations are internal to  $\varphi_E$ . Any apparent ontological gap between two domains (mind/matter, fact/value, natural/social) would require a discontinuity in  $\varphi_E$ , contradicting A10, or a causal influence crossing an ontological boundary, contradicting A11. Therefore no ontological gaps exist; all apparent discontinuities are gradient differences in informational-entropic structure within the continuous field. QED.

*Philosophical Significance: T8 dissolves the problems generated by ontological dualism in all its forms — including the Cartesian mind-body problem, the is-ought gap, and the gap between natural and social sciences.*

**T9 — The Sufficiency of the Entropic Field**

*Theorem: The Entropic Field, together with the Obidi Action, is metaphysically sufficient: no further ontological posit is required to account for any feature of reality.*

**Proof Sketch:**

By A12 (Sufficiency),  $\varphi_E$  and  $\mathcal{P}_{\text{Obidi}}$  together with the axioms are sufficient to ground all metaphysical facts. By T2, the Entropic Ground is unique. By A11, all causal relations are internal. Any proposed additional ontological posit (Platonic forms, theological entities, non-physical minds, etc.) would either (i) be internal to  $\varphi_E$  and thus already accommodated, or (ii) external and causally inert (by A11), contributing nothing to ontological explanation. By parsimony consistent with A12, the system is closed. QED.

*Philosophical Significance: T9 is the master result of the axiomatic system, establishing the self-sufficiency of Entropicity as a complete metaphysical framework.*

### **T10 — The Grounding of Value in Entropicity**

*Theorem: Value facts are real, objective features of  $\varphi_E$  as organized by high-order Entropic Subjects; they are not mere projections of subjective preference.*

#### **Proof Sketch:**

By A9, high-order Entropic Subjects are genuine ontological nodes within  $\varphi_E$ . By P6, value is a measure of informational coherence within  $\varphi_E$  as organized by such subjects. By D11, value is objectively grounded in the informational dynamics of the field. By A12, value facts are metaphysical facts groundable within the system. Since the ground of value is the objective informational structure of  $\varphi_E$  (not merely the subjective states of Entropic Subjects), value is objective in the requisite sense. QED.

*Philosophical Significance: T10 provides the metaphysical foundation for the ethical and aesthetic dimensions of Entropicity, offering a non-Platonic, non-theistic grounding for objective value.*

## **PART VI — COROLLARIES (C1-C8)**

---

*Corollaries follow from theorems with minimal additional argument. They represent the system's direct implications for long-standing philosophical problems.*

**C1 (from T5): The Abstraction of Substance.** Substance, in the sense of a fixed, self-subsistent, property-bearing substratum, is an abstraction from entropic processes. There are no substances in nature; there are only stable, coherent entropic trajectories that our cognitive architecture hypostatizes into substance-concepts for practical purposes. The ontological furniture of the universe consists not of things, but of processes and their relational structures.

---

**C2 (from T5): Personal Identity as Entropic Trajectory.** Personal identity is not a fixed, immutable essence (soul, ego-substance, Cartesian *res cogitans*) but an entropic trajectory within  $\varphi_E$ : a pattern of informational coherence that persists with sufficient continuity to constitute a numerically identical person across time. Identity admits of degrees; the discontinuities introduced by sleep, memory loss, or radical personality change are real but do not necessarily rupture identity, which requires only sufficient — not absolute — informational continuity.

---

**C3 (from T6): The Compatibility of Free Will and Entropicity.** Free will, properly understood, is the self-determining behavior of high-order Entropic Subjects: their capacity to determine their own trajectory within  $\varphi_E$  through the exercise of their own internal informational organization. Free will is not libertarian contra-causal freedom (which would violate A11) but is the highest expression of entropic self-organization: the degree of self-direction available to a system whose causal powers flow from its own high-order informational structure rather than from external entropic constraint alone.

---

**C4 (from T8, T9): Ontological Closure of the Universe.** The universe, understood as the totality of  $\varphi_E$ , is ontologically closed: there is no external ontological anchor, no realm of Forms beyond it, no God external to it, no noumenal realm behind it. If theological or Platonic realities exist, they do so

within the Entropic Field. The closure of the universe is not a limiting condition but its defining positive character as the total ontological ground.

---

**C5 (from T10): The Objectivity of Moral Facts.** Moral facts are grounded in the informational structure of  $\varphi_E$  as organized through high-order Entropic Subjects and their relational dynamics. Moral nihilism (no moral facts) and subjectivism (moral facts are merely personal preferences) are both false within Entropicity. Moral realism is true in the sense that value is an objective feature of the field — though it is not Platonic (not mind-independent in a transcendent sense) but is constituted through the real dynamics of high-order entropic organization.

---

**C6 (from T6, T7): Mind-Matter Monism.** Mind and matter are not two substances (Descartes), nor is mind reducible to matter (eliminative materialism), nor matter to mind (idealism). They are two aspects — two organizational modes — of the same Entropic Field, differentiated by their degree of self-referential informational organization. The "hard problem" of consciousness dissolves in Entropicity: it is not a problem of explaining how mind arises from a fundamentally non-mental matter, but of understanding how  $\varphi_E$  organizes itself into the self-referential configurations that constitute conscious experience.

---

**C7 (from A4, P1): Probabilistic Entropicity.** The cosmos is not deterministic in the classical Laplacean sense, but probabilistically entropic: the Obidi Action extremizes over a probability distribution across configuration space, not a single determined trajectory. The future is genuinely open at the level of  $\varphi_E$ ; quantum indeterminacy is a domain-specific manifestation of this fundamental ontological probability. Entropicity is therefore compatible with genuine novelty, creativity, and ontological openness.

---

**C8 (from A10, D2): Death as Entropic Transition.** Death, for an Entropic Particular, is a radical entropic transition — the dissolution of a coherent informational trajectory into the broader Entropic Field — but not ontological annihilation. By T1 (Impossibility of Absolute Non-Being), the information constituting a person cannot be absolutely destroyed; it is redistributed within  $\varphi_E$ . Whether this redistribution constitutes any meaningful form of personal continuity is an open problem (identified in Part VII), but the claim that death is total annihilation is inconsistent with the axioms of Entropicity.

---

## **PART VII — OPEN PROBLEMS AND FUTURE AXIOMATIZATION**

---

*The Axioms of Entropicity identify several foundational problems that the current system does not yet fully resolve. These are not objections to the system but its productive frontier: the regions where the axiomatic framework opens into further inquiry.*

**OP1. The Measure Problem of Entropic Possibility-Space.** The axioms posit that the Obidi Action extremizes over the configuration space of  $\varphi_E$ , but do not specify the correct probability measure over that space. What is the natural measure of entropic possibility? This is the metaphysical analogue of the quantum cosmological measure problem, and its resolution is essential for a fully quantitative Entropicity.

**OP2. The Informational Boundary Conditions of the Entropic Field.**

What are the initial (or boundary) conditions of  $\varphi_E$ ? Is the field eternal, or does it have a beginning? If it has a beginning, what grounds the initial informational configuration? The system's axioms are silent on cosmological origins, and a future axiomatization must address this through a theory of entropic initial conditions.

**OP3. The Precise Form of the Obidi Action.** The Obidi Action is posited as the fundamental operator of entropic actualization, but its precise mathematical form remains to be specified. Future formalization must construct the explicit functional form of  $\mathcal{S}_{\text{Obidi}}$  such that known physical laws (quantum mechanics, general relativity, thermodynamics) are derivable as constrained instances at appropriate levels of  $\varphi_E$ .

**OP4. The Continuity-Discreteness Problem.** A10 asserts the ontological continuity of  $\varphi_E$ , yet quantum mechanics and quantum field theory exhibit fundamental discreteness (quantization). The precise account of how discrete quantum phenomena emerge from a continuous ontological field without contradiction requires a future axiomatization of the quantum level within Entropicity — a Quantum Axiomatics of Entropicity.

**OP5. The Metric of Post-Mortem Informational Persistence.** C8 establishes that death is an entropic transition and not absolute annihilation, but leaves open the question of whether the informational redistribution following death constitutes any form of meaningful personal continuity. A future axiomatization of personal identity within Entropicity must define formal criteria for whether information redistribution preserves or constitutes personal identity in a morally and metaphysically relevant sense.

**OP6. The Modal Logic of Entropicity.** The definitions of necessity (D12) and contingency (D13) in terms of "all possible configurations" and "some configurations" of  $\varphi_E$  require a precise modal logic. The entropic modal logic of the system — specifying the accessibility relations between possible configurations of  $\varphi_E$ , the structure of entropic possible worlds, and the interaction between modality and time — is a major open formal problem.

**OP7. The Ethical Axiomatization of Entropicity.** P6 and T10 establish the grounding of value in entropic informational coherence, but the system does not yet contain a formal ethical axiomatization: a set of ethical axioms derivable from

the metaphysical axioms that generates a systematic moral theory. The construction of the Ethical Axioms of Entropicity — addressing obligation, rights, justice, virtue, and moral progress — is a priority for the next stage of the system's development.

**OP8. Inter-Systemic Comparison and Falsifiability.** A formal axiomatic metaphysical system must be testable, at least indirectly, through its empirical implications and through formal comparison with competing systems. A rigorous methodology for the inter-systemic comparison of Entropicity with physicalism, panpsychism, idealism, process philosophy, and structural realism — and for identifying the observational and formal consequences that would confirm or disconfirm Entropicity relative to these rivals — remains to be developed.

## Directions for Future Formalization

- **Modal Logic of Entropicity:** Develop a formal entropic modal logic (S5-Entropicity or equivalent) specifying accessibility between entropic configurations.
- **Quantum Axiomatics:** Construct a sub-system of axioms governing the quantum level of  $\varphi_E$ , deriving quantum indeterminacy and entanglement as entropic phenomena.
- **Ethical Axiomatization:** Extend the system with formal ethical axioms grounded in P6 and T10, generating a systematic Entropicity-based moral theory.
- **Statistical Gravity Formalism:** Develop the full mathematical apparatus of Statistical Gravity as the informational-geometric grounding of general relativity within the Entropic Field.
- **Entropic Theory of Meaning:** Axiomatize the semantics of the system itself — accounting for how Entropic Subjects generate, transmit, and interpret meaning within  $\varphi_E$ .

# APPENDIX — NOTATION SUMMARY

*The following table provides a complete reference for all symbols, formal terms, and their meanings within the axiomatic system of Entropicity.*

| Symbol / Term                | Name               | Meaning within Entropicity                                                                                                                          |
|------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi_E$                  | The Entropic Field | Ontological ground of all being; the total informational medium from which all particular entities, processes, and relations emerge (Primitive I.1) |
| $\mathcal{I}$                | Information        | The fundamental currency of ontological structure; the primitive differentiating factor constituting all being (Primitive I.2)                      |
| S                            | Entropy            | The measure of informational potentiality within a region or state of $\varphi_E$ ; quantified formally by Shannon entropy H (Primitive I.3; P1)    |
| $\mathfrak{B}$               | Becoming           | The irreversible ontological process by which being actualizes through entropic increase (Primitive I.4)                                            |
| $\Sigma$                     | Entropic Subject   | A self-organizing, self-referential informational node within $\varphi_E$ ; the ontological basis of consciousness and selfhood (Primitive I.5)     |
| $\mathcal{S}_{\text{Obidi}}$ | The Obidi Action   | The fundamental operator governing entropic actualization; the master law from which all domain-specific laws are                                   |

| Symbol / Term               | Name                    | Meaning within Entropicity                                                                                                                     |
|-----------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
|                             |                         | derived (Primitive I.6)                                                                                                                        |
| $\tau(E)$                   | Entropic Trajectory     | The unique informational-entropic path of an Entropic Particular E through $\varphi_E$ ; constitutes its identity (D3, D9)                     |
| $\mathcal{J}(\omega) > 0$   | Being Condition         | The necessary and sufficient condition for a state $\omega$ of $\varphi_E$ to qualify as a being (D1)                                          |
| $\mathcal{J} \rightarrow 0$ | Non-Being Limit         | The asymptotic limit approached by Non-Being; never fully realized (D2, T1)                                                                    |
| $R(E_1, \dots, E_n)$        | Relation                | A structural dependency between Entropic Particulars, mediated by $\varphi_E$ (D4)                                                             |
| $dS/dt \geq 0$              | Entropic Law            | The formal expression of A3 and A5: entropy does not decrease; Becoming is irreversible (A3, A5)                                               |
| $\mathbb{G}(\varphi_E)$     | Informational Geometry  | The geometric structure of the Entropic Field induced by entropy distributions; the topology of reality (A7, D7)                               |
| $E_H$                       | Higher-Order Particular | An Entropic Particular with irreducibly emergent properties arising from the collective dynamics of lower-order Entropic Particulars (A8, D10) |
| $H = -\sum p_i \log p_i$    | Shannon Entropy         | The operative formal measure of ontological potentiality within $\varphi_E$ , adopted by P1                                                    |
| V                           | Value Measure           | A measure of informational coherence and complexity within an Entropic Subject; the formal correlate of ethical and aesthetic value (D11,      |

| Symbol / Term | Name                  | Meaning within Entropicity                                                                                                 |
|---------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------|
|               |                       | P6, T10)                                                                                                                   |
| A1-A12        | Axioms                | The twelve foundational assertions of Entropicity, accepted without proof within the system                                |
| D1-D15        | Definitions           | Formal definitions of the central concepts of Entropicity, constructed from primitive concepts                             |
| P1-P6         | Postulates            | Domain-specific assumptions extending the axioms to particular areas of inquiry                                            |
| T1-T10        | Theorems              | Propositions derived by valid inference from axioms, postulates, and prior theorems                                        |
| C1-C8         | Corollaries           | Propositions following immediately from theorems with minimal additional argument                                          |
| OP1-OP8       | Open Problems         | Foundational problems identified but not yet resolved by the current axiomatic system                                      |
| ToE           | Theory of Entropicity | The full metaphysical system of Obidi's Metaphysics, of which this axiomatic document is the foundational formal statement |

### Closing Note on Axiomatization

This document represents the first formal axiomatic edition of the Theory of Entropicity. It is not claimed to be final or complete. An axiomatic system is a living structure: open to refinement of its axioms

in response to internal inconsistencies discovered by formal analysis, to revision of postulates in response to developments in empirical science and formal logic, and to extension through new theorems and corollaries as the system matures.

The measure of the system is not whether its axioms are self-evidently true — no foundational axioms are — but whether the system as a whole is consistent, explanatorily powerful, ontologically economical, and capable of generating a progressive and expanding research program in metaphysics, philosophy of mind, ethics, and the foundations of science.

---

**Author:** John Onimisi Obidi (INFINITE)

**System:** Obidi's Metaphysics — Theory of Entropicity

**Date of First Formal Axiomatization:** June 10, 2026

**Location:** Regina, Saskatchewan, Canada

*"To axiomatize is to take thought seriously enough to hold it accountable."*

— INFINITE

---

© 2026 John Onimisi Obidi (INFINITE). Obidi's Metaphysics — Theory of Entropicity.

First Formal Axiomatic Edition. All rights reserved.

*This document constitutes the foundational axiomatic record of the Theory of Entropicity.*